**Baseball Team Management & LoRa Play-Calling App**

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# 1. Project Overview

This project delivers a **baseball-only team management and scoring app**, similar to GameChanger, with an NFHS-compliant **one-way electronic play-calling system** using **LoRa technology**.

The system enables coaches to send **offensive signals, defensive alignments, and pitch calls** from a mobile app to **player-worn watches**, while strictly adhering to NFHS rules regarding electronic devices and in-game communication.

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# 2. NFHS Compliance Summary (High-Level)

The app is designed to comply with NFHS baseball rules by enforcing:

- **One-way communication only** (coach → player)
- **No player input or responses**
- **No internet or cellular dependency during play**
- **No statistical, analytical, or video data transmitted to players**
- **Use limited to permitted purposes** (signs, pitch calls, positioning)

All compliance safeguards are built directly into the system architecture.

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# 3. Objectives

- Provide NFHS-legal electronic sign replacement
- Prevent sign stealing while remaining rule-compliant
- Improve communication clarity without competitive imbalance

- Offer reliable functionality without cellular service
- Support standard baseball scoring and team management

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# 4. Target Users

- NFHS high school baseball programs
- Competitive youth leagues following NFHS rules
- Coaches and assistant coaches
- Players equipped with approved wearable devices
- Parents and fans (view-only access)

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# 5. Core System Components

## 5.1 Coach Mobile App

- Play selection and transmission
- Game scoring (coach-only)
- Roster and lineup management
- Pitch calling interface
- LoRa transmitter control
- **NFHS compliance enforcement logic**

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## 5.2 Player Watch App (NFHS Mode)

- Receives **one-way LoRa messages only**
- Displays simple play instructions
- No buttons, reply options, or input
- Automatic clearing of plays
- Lockout during dead-ball restrictions if required

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### 5.3 Backend Platform

- Account management and permissions
- Game data storage (coach/fan use only)
- Compliance logging
- Device pairing management

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# 6. Key Features

## 6.1 Game Scoring & Stats (Coach / Fan Only)

- Pitch-by-pitch scoring
- Box scores and summaries
- Season stats and player histories

#### NFHS Safeguard

- No live stats or analytics are sent to players
- No real-time performance data accessible during play

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### 6.2 Team & Roster Management

- Player profiles
- Lineups and defensive positioning
- Pitch count tracking
- Practice and game schedules

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# 7. NFHS-Compliant LoRa Play Calling (Core Feature)

## 7.1 Allowed Play Types

**Offensive**

- Steal / no steal
- Bunt / swing
- Hit-and-run
- Take signs

**Defensive**

- Defensive alignments
- Shift instructions
- Pickoff coverage
- Bunt defense positioning

**Pitch Calling**

- Pitch type

- Location zone
- Sequence number (optional)

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### 7.2 NFHS Technical Restrictions (Enforced)

| Requirement | Implementation |
|---|---|
| One-way communication | LoRa is transmit-only to watches |
| No player input | Watch UI has no controls |
| No internet dependency | LoRa functions offline |
| No analytics to players | Messages limited to pre-defined codes |
| Coach-only control | Role-based permissions |
| No audio | Haptic + visual only |

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## 8. Wearable (Watch) Experience

- Icon-based or coded display
- Color-coded by play type
- Haptic vibration for alerts
- Automatic message expiration
- No storage of historical plays
- Screen locks after display timeout

**NFHS Safeguard**

- Watch cannot browse, review, or replay prior calls

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# 9. Connectivity & Technology

## 9.1 LoRa Communication

- Long-range, low-power transmission
- Field-level base station or coach device
- Encrypted payloads
- Team- and game-specific pairing
- No external network access during games

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## 9.2 Technology Stack (Proposed)

- Mobile: iOS / Android
- Watch: Custom LoRa firmware or compliant smartwatch
- Backend: Node.js / Django
- Database: PostgreSQL
- Cloud hosting: AWS / GCP

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# 10. User Roles & Permissions

- **Head Coach** – Full control
- **Assistant Coach** – Limited play calling
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- **Scorer -** Able to record plays after they happen
- **Player** – Passive receiver only

- **Parent/Fan** – View-only (outside live play communication)

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# 11. Security & Rule Compliance

- End-to-end encrypted LoRa messages
- Device pairing per game
- NFHS compliance mode (cannot be disabled during NFHS games)
- Audit logs for play transmissions
- Administrative lockout for non-compliant features

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# 12. Development Phases

### Phase 1: NFHS MVP

- Baseball scoring
- Roster management
- One-way LoRa play transmission
- Watch display

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### Phase 2: Compliance Hardening

- NFHS lock modes
- Audit logging
- Umpire/AD compliance documentation

- Device certification workflows

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## 13. Success Metrics

- 100% one-way message delivery
- Zero NFHS compliance violations
- Coach adoption rate
- Reduced sign confusion
- Game-day reliability